NCO · FOUR-ASSET WALKTHROUGH

Nested Clustered Optimization

A step-by-step four-asset example

Two stocks and two bonds, from a noisy correlation matrix to final weights

Denoise and cluster → optimize inside each cluster → shrink to clusters → optimize across them, computed by hand, then rebuilt with Claude Code.

NCO confines optimizer noise to small clusters

Nested Clustered Optimization was proposed by Marcos López de Prado (2019).

1

Why: inverting a large noisy matrix is fragile

Mean-variance optimization lets errors in one corner of the matrix move every weight.

2

How: solve small problems, then one problem across clusters

Each cluster is optimized on its own and then treated as a single synthetic asset.

3

Result: the same optimizer, much smaller weight errors

Our example ends at A 0.112, B 0.034, C 0.687, D 0.167.

Every number in this deck is computed by hand on a two-stock, two-bond example

Six parts take us from basics to a hands-on build

Parts 2 and 3 are the four NCO steps; the same four-asset example runs throughout.

Part	Topic	What you will see
01	Basics	Why MVO is fragile, noise eigenvalues, clusters, the example
02	Step 1	Denoise the correlation matrix and find the clusters (ONC)
03	Steps 2–4	Optimize inside, shrink to clusters, optimize across
04	Results	A what-if, a simulation, NCO versus HRP and MVO
05	Build	Rebuild everything with Claude Code prompts
	Wrap-up	Three lines to remember

01

PART 1 · BASICS

Background for Reading NCO

Mean-variance weights, eigenvalue noise, clusters and silhouettes, and the example.

Minimum variance needs the inverse covariance

This one formula is used twice inside NCO: within each cluster and across clusters.

Minimum-variance weights (weights sum to 1)

$$w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

Symbols

- Σ: covariance matrix
- 1: a column of ones

Why it breaks on large problems

- Inverse blows up small eigenvalues
- Small eigenvalues are mostly noise
- Highly correlated assets → huge long-short bets
- One bad corner moves every weight

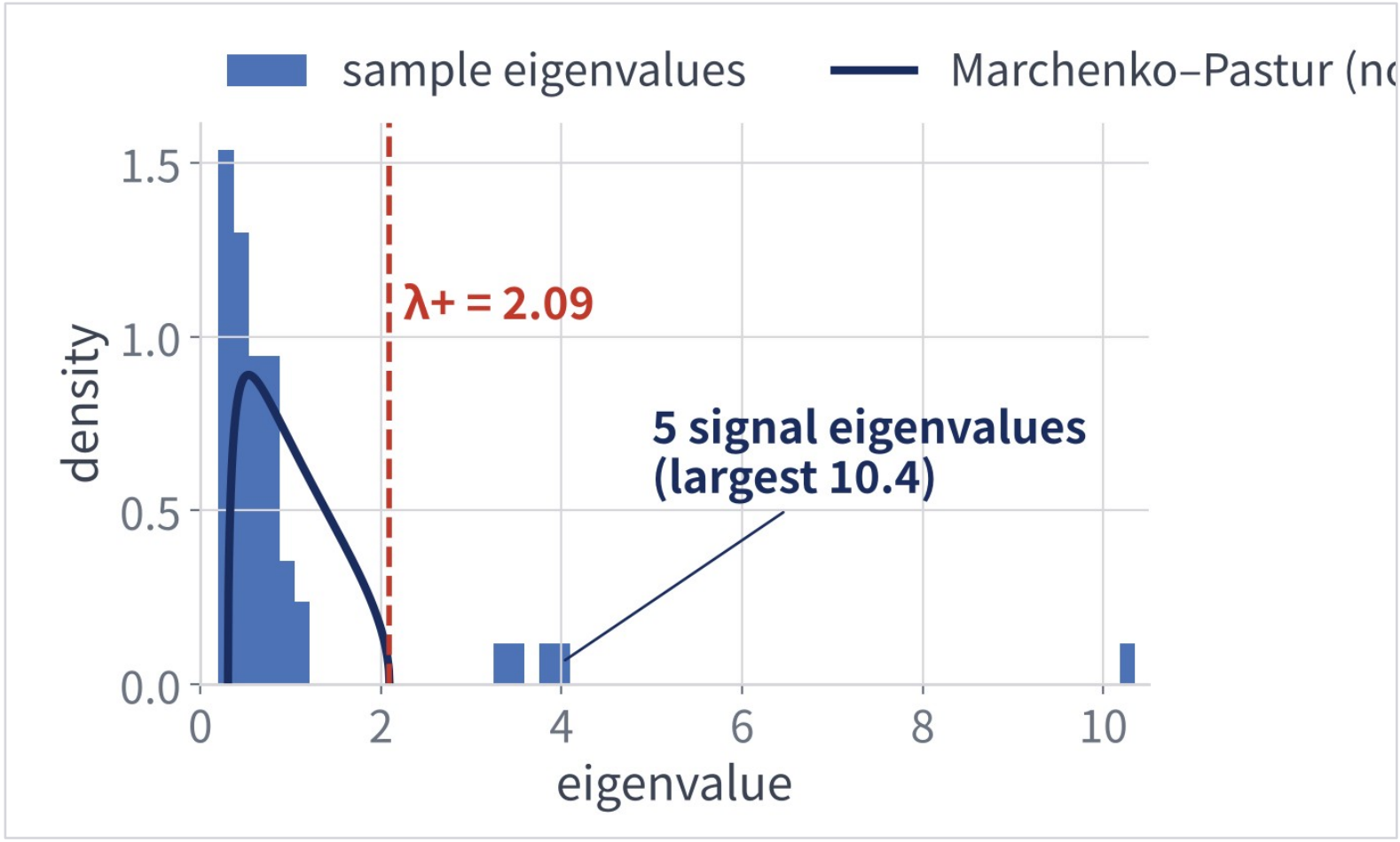
López de Prado calls this instability Markowitz's curse

Most eigenvalues of a sample correlation are noise

Simulation: 50 assets in 5 groups, 250 daily returns. The curve is what pure noise would produce.

Reading it

- 45 sit under the curve
- 5 stand far right
- Those 5 are structure
- Step 1 keeps only them



Silhouettes measure how well an asset fits

a = average distance to its own cluster; b = average distance to the nearest other cluster.

Correlation distance (same as in HRP)

$$\text{distance} = \sqrt{\frac{1-\text{corr}}{2}}$$

Silhouette of asset i

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

How to read s

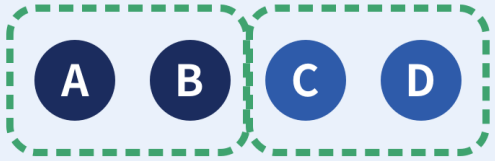
- Near 1: snug in its own cluster
- Near 0: on the border
- Negative: in the wrong cluster
- Step 1 uses s to choose K

A good clustering makes every asset's silhouette high and similar

NCO works in four steps, from small to large

Steps 2 to 4 reuse the same minimum-variance formula; only the size of the problem changes.

1 Denoise + cluster



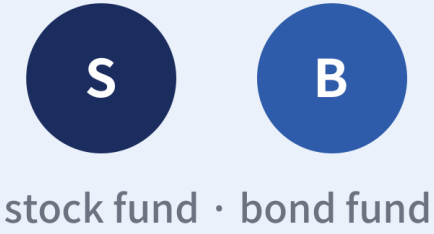
• clean the correlation
ONC picks K clusters

2 Inside each



• min-variance in
each small cluster

3 Shrink to K



• each cluster becomes
one synthetic asset

4 Across clusters



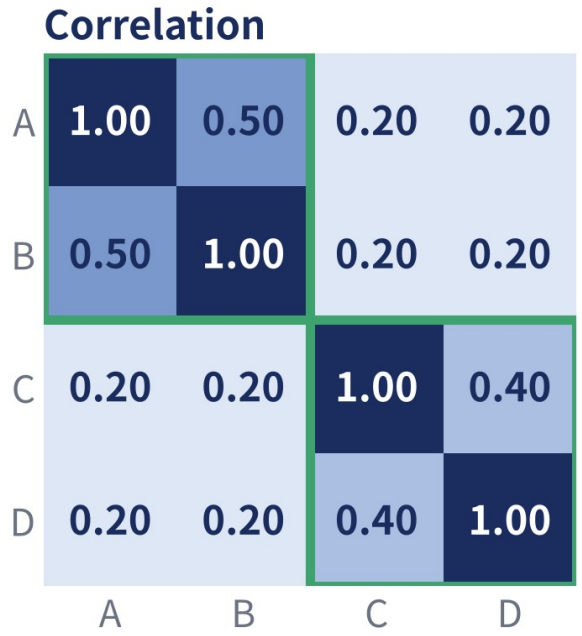
• min-variance on K
assets, then multiply

The example has two stocks and two bonds

Made-up numbers. Stocks correlate 0.5, bonds 0.4, and every stock–bond pair 0.2.

Asset	Type	Volatility σ	Variance σ²
A	stock	0.15	0.0225
B	stock	0.20	0.0400
C	bond	0.08	0.0064
D	bond	0.12	0.0144

Pair	Covariance
A–B	0.0150
C–D	0.0038
stock–bond	0.0024 to 0.0048



Covariance = correlation × σ × σ, e.g. A–B = 0.5 × 0.15 × 0.20 = 0.0150

02

PART 2 · STEP 1

Denoising and Clustering

Flatten the noise eigenvalues, then let the silhouette score choose the clusters.

Eigenvalues below the MP bound count as noise

N assets, T observations; $\sigma^2 = 1$ for a correlation matrix. Below: the simulation, N = 50 and T = 250.

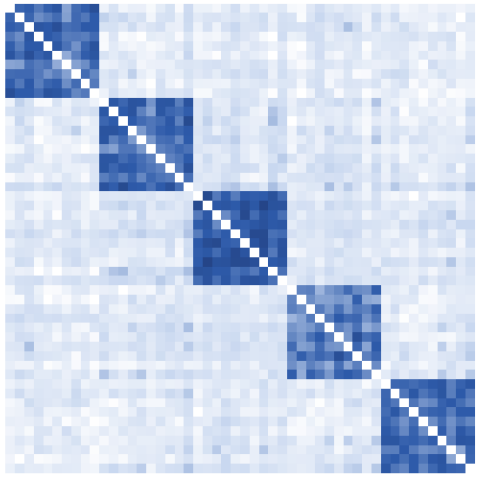
Largest eigenvalue pure noise can make

$$\lambda_+ = \sigma^2 \left(1 + \sqrt{N/T}\right)^2$$

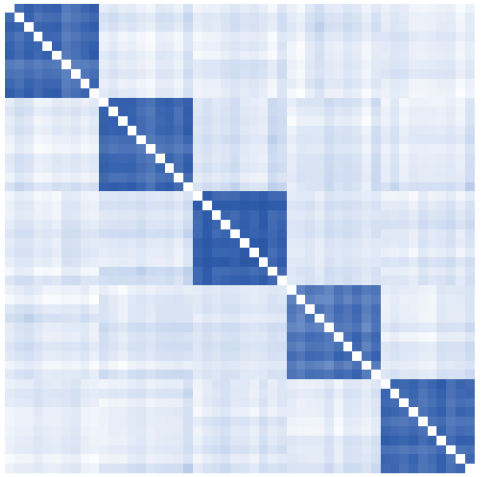
$$\lambda_+ = \left(1 + \sqrt{50/250}\right)^2 \approx 2.09$$

Fewer observations per asset → higher bound → more noise.

Sample correlation (noisy)



Denoised correlation



After cleaning, the five group blocks stand out and the speckle between them fades

Denoising keeps signal and averages the noise

Constant residual eigenvalue method, López de Prado (2020).

Decompose

eigenvalues

Split the correlation matrix into eigenvalues and eigenvectors.

Sort

signal vs noise

Eigenvalues above λ+ are signal; the rest are noise.

Flatten

average

Replace every noise eigenvalue by their average, so the total is unchanged.

Rebuild

rescale

Rebuild the matrix and rescale its diagonal back to 1.

The matrix keeps its size and its trace; only the noisy directions are calmed

ONC tries several K and keeps the best score

ONC = Optimal Number of Clusters: K-means on correlation distances, scored by silhouettes.

Score of one clustering

$$\text{quality} = \frac{\text{mean}(s)}{\text{std}(s)}$$

Why mean over std

- High mean: assets fit well
- Low std: no misfits

The ONC loop

- Turn correlations into distances
- For K = 2, 3, ...: run K-means
- Score each result
- Keep the K with the highest score

The number of clusters is chosen by the data, not by the analyst

Within-cluster distances are the shortest

We take the four-asset correlation as already clean; with four assets there is no noise left to flatten.



Rule

$$\text{distance} = \sqrt{\frac{1-\text{corr}}{2}}$$

Our numbers

$$d_{AB} = \sqrt{(1-0.5)/2} = 0.500$$
$$d_{CD} = \sqrt{(1-0.4)/2} = 0.548$$
$$d_{\text{cross}} = \sqrt{(1-0.2)/2} = 0.632$$

A–B (0.500) and C–D (0.548) are closer than any stock–bond pair (0.632)

Two clusters score far better than three

Silhouettes use the correlation distances above. A single-asset cluster gets s = 0.

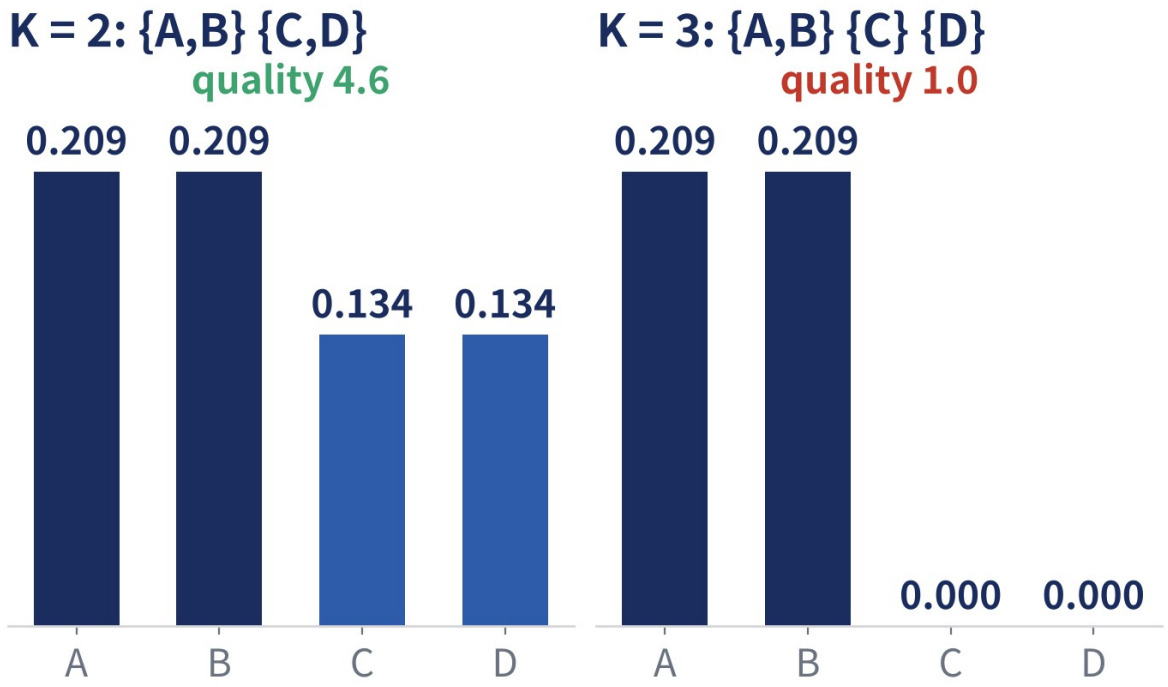
Silhouettes for K = 2

$$s_A = s_B = \frac{0.632 - 0.500}{0.632} = 0.209$$

$$s_C = s_D = \frac{0.632 - 0.548}{0.632} = 0.134$$

Scores

$$K = 2: \ \frac{0.172}{0.038} = 4.6 \qquad K = 3: \ \frac{0.105}{0.105} = 1.0$$



ONC returns two clusters: stocks {A, B} and bonds {C, D}

03

PART 3 · STEPS 2–4

Optimizing Inside, Then Across

Minimum variance inside each cluster, shrink each cluster to one asset, minimum variance across.

Inside the stock cluster, minimum variance favours A

The same formula as in Basics, written out for two assets.



Two-asset minimum variance

$$w_1 = \frac{\mathrm{Var}_2 - \mathrm{Cov}_{12}}{\mathrm{Var}_1 + \mathrm{Var}_2 - 2\,\mathrm{Cov}_{12}}$$

Stocks A and B

$$w_A = \frac{0.0400 - 0.0150}{0.0225 + 0.0400 - 2(0.0150)} = 0.769$$

$$w_B = 1 - 0.769 = 0.231$$

A is calmer (variance 0.0225 vs 0.0400), so it takes 76.9% of the stock cluster

Inside the bond cluster, minimum variance favours C

Only the bonds' own 2 × 2 covariance is used; stocks play no part here.



Bonds C and D

$$w_C = \frac{0.0144 - 0.0038}{0.0064 + 0.0144 - 2(0.0038)} = 0.805$$

$$w_D = 1 - 0.805 = 0.195$$

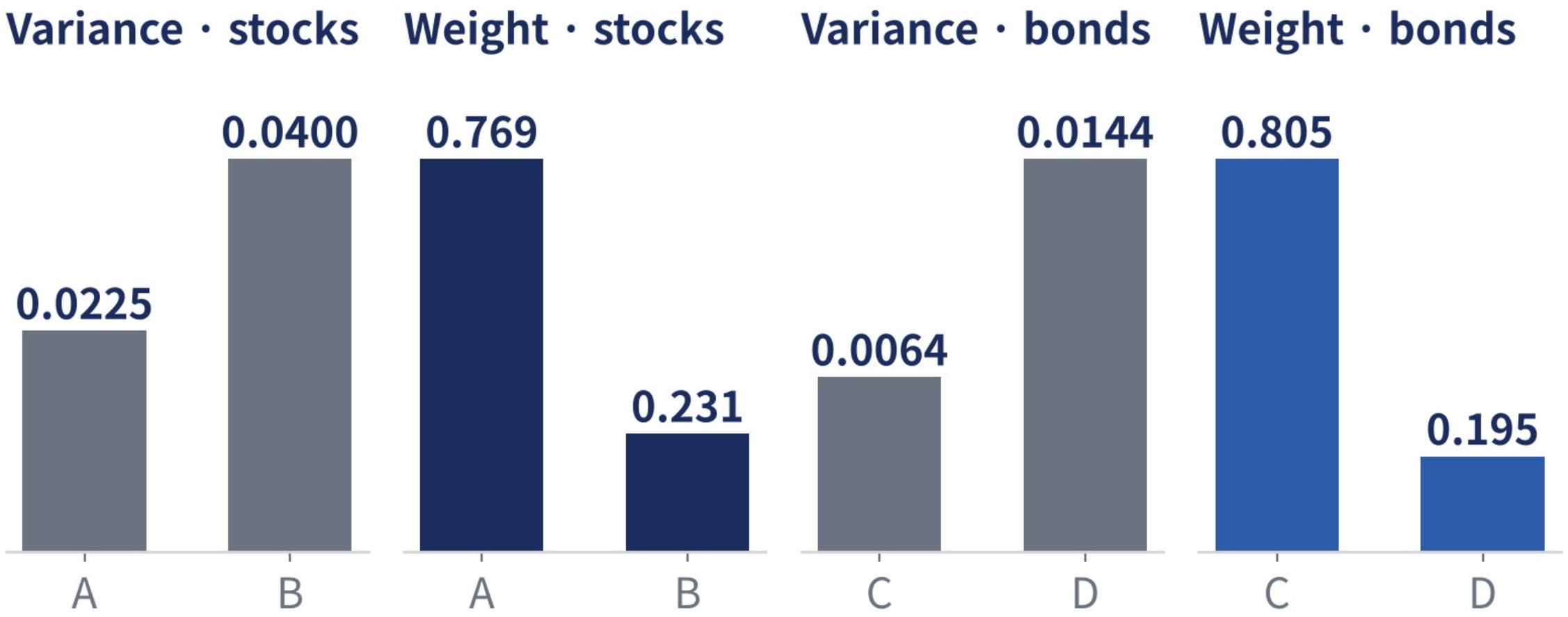
Notice

- Two small inverses, never a 4 × 4

C takes 80.5% of the bond cluster

The calmer asset in each cluster gets the larger share

Correlation matters too: A gets 0.769, more than inverse variance alone would give (0.640).



Each cluster becomes one synthetic asset

Plug the Step 2 weights into each cluster's 2 × 2 covariance.



$$\text{Var} = w' \Sigma w$$

Stock fund

$$\text{Var}_{\text{stock}} = 0.769^2(0.0225) + 0.231^2(0.040) + 2(0.769)(0.231)(0.015) = 0.0208$$

Bond fund

$$\text{Var}_{\text{bond}} = 0.805^2(0.0064) + 0.195^2(0.0144) + 2(0.805)(0.195)(0.0038) = 0.0059$$

The stock fund (0.0208) is about 3.5 times as risky as the bond fund (0.0059)

W shrinks four assets into a 2 × 2 matrix

Each column of W holds one cluster's weights and zeros elsewhere.

Reduced covariance

$$\hat{\Sigma} = W' \Sigma W$$

Off-diagonal: stock–bond

$$\begin{aligned} \mathrm{Cov}_{\mathrm{stock,bond}} &= w'_{\mathrm{stock}} \, \Sigma_{\mathrm{cross}} \, w_{\mathrm{bond}} \\ &= 0.0028 \end{aligned}$$

The only place where the four cross covariances enter.

W: one column per cluster

	stock	bond
A	0.769	0
B	0.231	0
C	0	0.805
D	0	0.195

W′ Σ W

Reduced covariance (2 × 2)

	stock fund	bond fund
stock fund	0.0208	0.0028
bond fund	0.0028	0.0059

Four assets become two; the next optimization is a 2 × 2 problem

Across clusters, the bond fund takes 85%

The same two-asset formula, now applied to the stock fund and the bond fund.

A	B	C	D
Var 0.0225	Var 0.0400	Var 0.0064	Var 0.0144
Stocks {A, B}		Bonds {C, D}	
w=0.769	w=0.231	w=0.805	w=0.195
Var 0.0208	Cov 0.0028	Var 0.0059	
ω = 0.146		ω = 0.854	
. . .	. . .	. . .	. . .

$$\hat{\Sigma} = \begin{pmatrix} 0.0208 & 0.0028 \\ 0.0028 & 0.0059 \end{pmatrix}$$

Cluster weights ω

$$\omega_{\text{stock}} = \frac{0.0059 - 0.0028}{0.0208 + 0.0059 - 2(0.0028)} = 0.146$$

$$\omega_{\text{bond}} = 1 - 0.146 = 0.854$$

The calmer bond fund takes 85.4% of the portfolio

Step 4 can use inverse variance instead

López de Prado uses the optimizer at both levels; inverse variance is a simpler variant.

Minimum variance (uses Cov 0.0028)

$$\omega_{\text{stock}} = \frac{0.0059 - 0.0028}{0.0208 + 0.0059 - 2(0.0028)} = 0.146$$

$$\omega_{\text{bond}} = 1 - 0.146 = 0.854$$

Inverse variance (ignores Cov)

$$\omega_{\text{stock}} = \frac{1/0.0208}{1/0.0208 + 1/0.0059} = 0.221$$

Step 4 rule	Stock fund	Bond fund	Uses correlation?
Minimum variance	0.146	0.854	yes
Inverse variance	0.221	0.779	no

A positive stock–bond covariance pushes more weight to bonds

Final weights multiply the two levels together

Each asset's weight = its share inside the cluster × the cluster's share.



$$w^* = W\,\omega$$

Our numbers

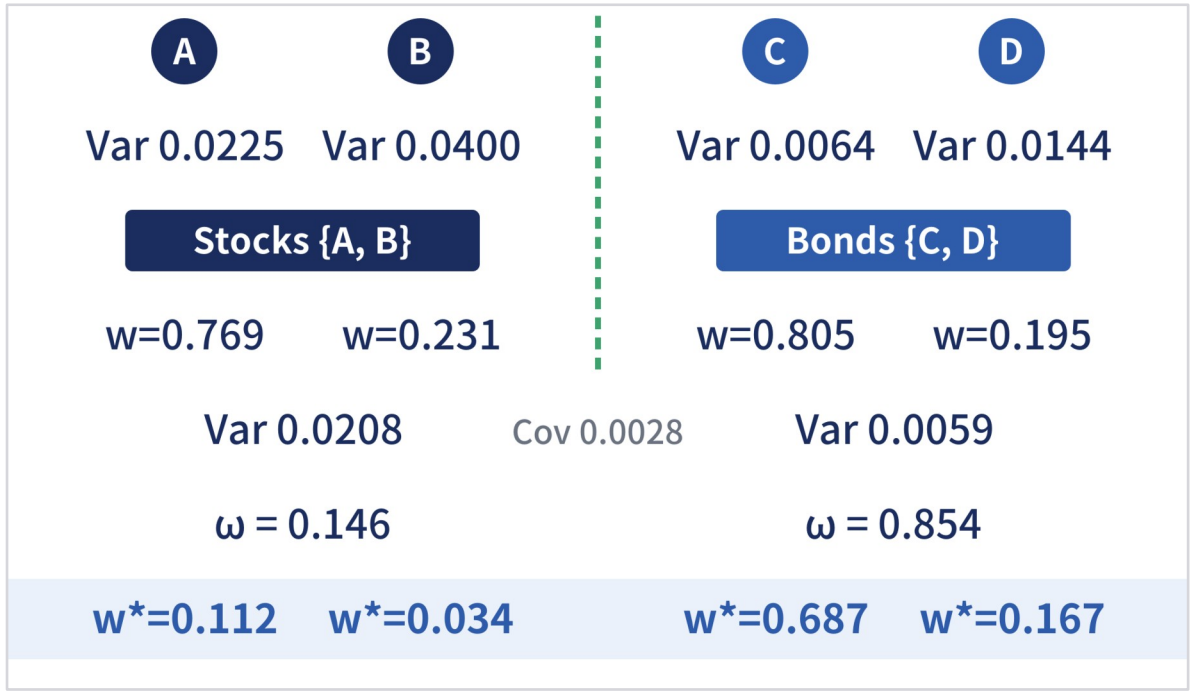
$$w_A = 0.769 \times 0.146 = 0.112$$
$$w_B = 0.231 \times 0.146 = 0.034$$
$$w_C = 0.805 \times 0.854 = 0.687$$
$$w_D = 0.195 \times 0.854 = 0.167$$

$$0.112 + 0.034 + 0.687 + 0.167 = 1.000$$

Final weights: A 0.112, B 0.034, C 0.687, D 0.167

Read top to bottom, NCO fits in one picture

Columns are the assets; the blue row is the answer. Compare with full MVO on the same matrix.



	A	B	C	D
NCO	0.112	0.034	0.687	0.167
Full MVO	0.135	0.011	0.701	0.153

Why they are close

- Clean structure, no noise
- NCO's gain shows with noise

On a clean matrix NCO lands near MVO; noise makes the difference

04

PART 4 · RESULTS

Reading the Results

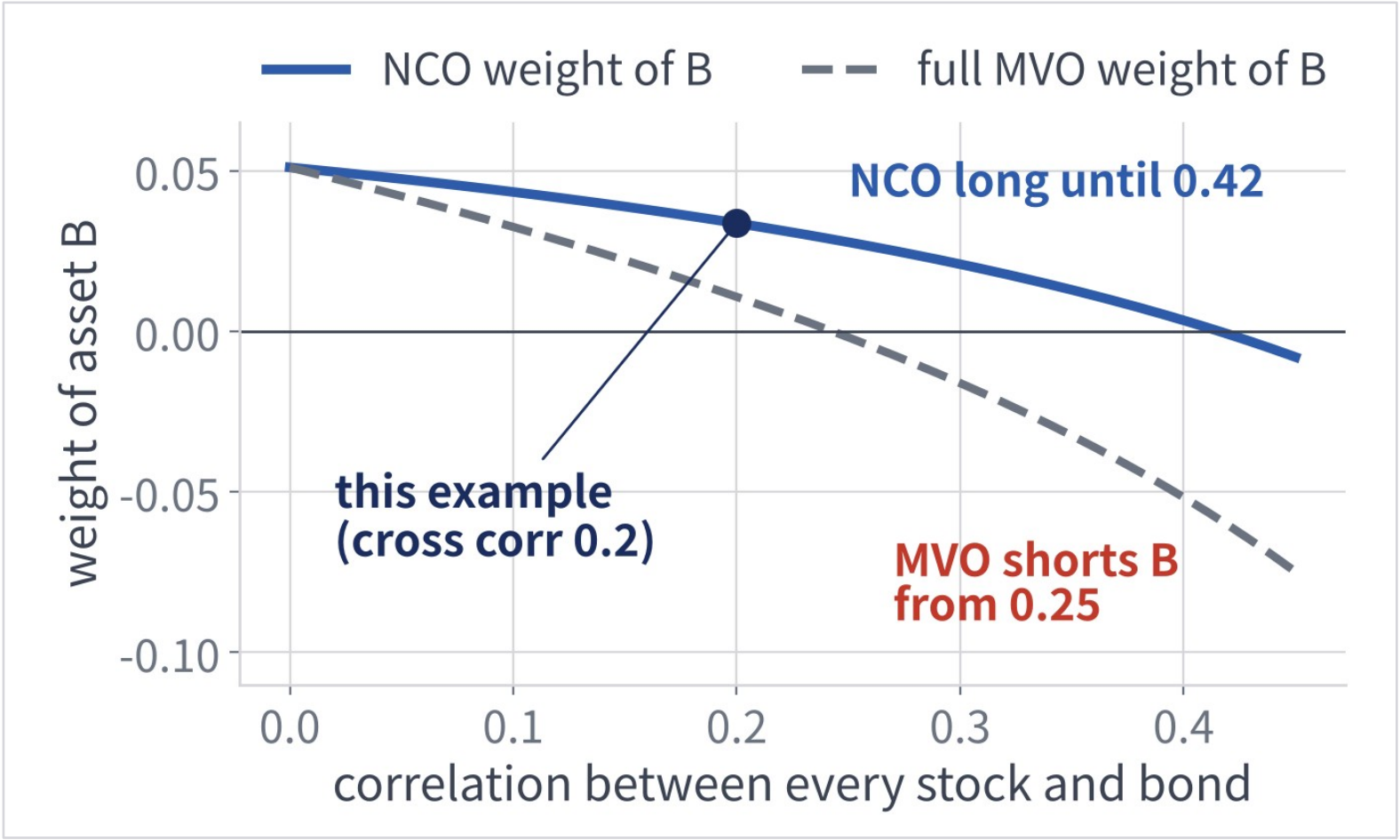
A what-if on correlation, a simulation of weight errors, and how NCO compares with HRP.

Full MVO shorts B much sooner than NCO

Only the stock–bond correlation changes; everything else stays as in the example (illustration).

Why

- MVO: B short from 0.25
- NCO: B short only at 0.42
- B's cluster share is fixed
- Noise stays in its cluster

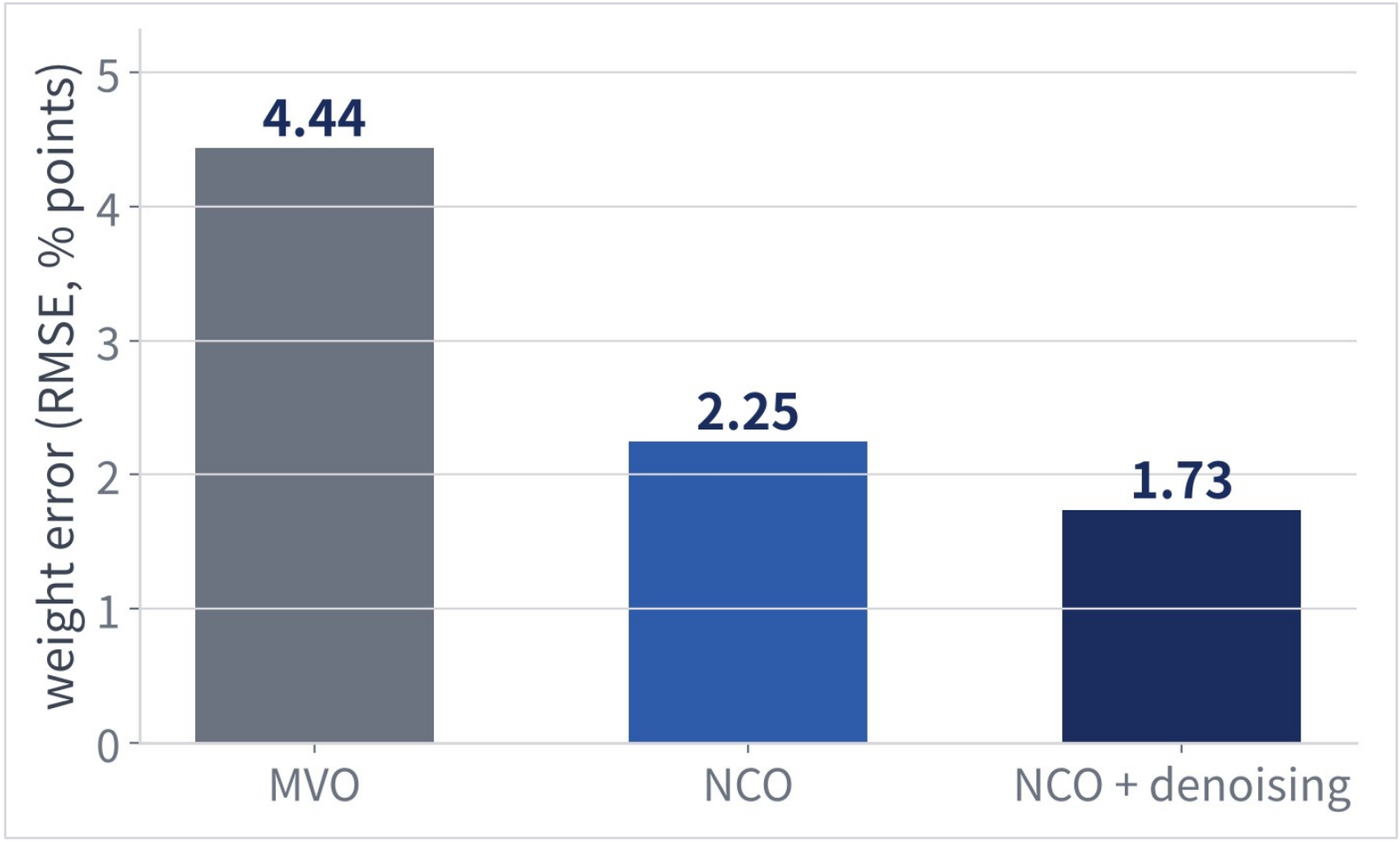


NCO halves the weight error of MVO

40 assets in 4 groups, 120 observations, 300 trials; error vs the true weights (illustration).

Reading it

- MVO: 4.44 points
- NCO: 2.25 points
- With denoising: 1.73
- Same data, same optimizer



NCO sits between MVO and HRP

HRP never inverts; MVO inverts everything; NCO inverts only small blocks.

	MVO	HRP	NCO
Matrix inverse	whole N × N	never	each cluster + K × K
Structure	none	tree, split in halves	clusters chosen by ONC
Expected returns	can use	not used	can use (max Sharpe)
Short positions	common	none	possible
Main risk	noise everywhere	ignores cross-group covariance	wrong clusters

Without return forecasts use HRP; for a stable optimizer use NCO

NCO buys stability at the cost of a clustering step

Points from López de Prado (2019, 2020) and practice since.

What NCO does well

- Keeps the optimizer
- Noise stays inside clusters
- Works with any objective
- Fits naturally with denoising

What to watch

- Results hinge on the clusters
- K-means differs run to run
- Short positions still possible
- Add constraints for long-only

Check cluster stability before trusting the weights

05

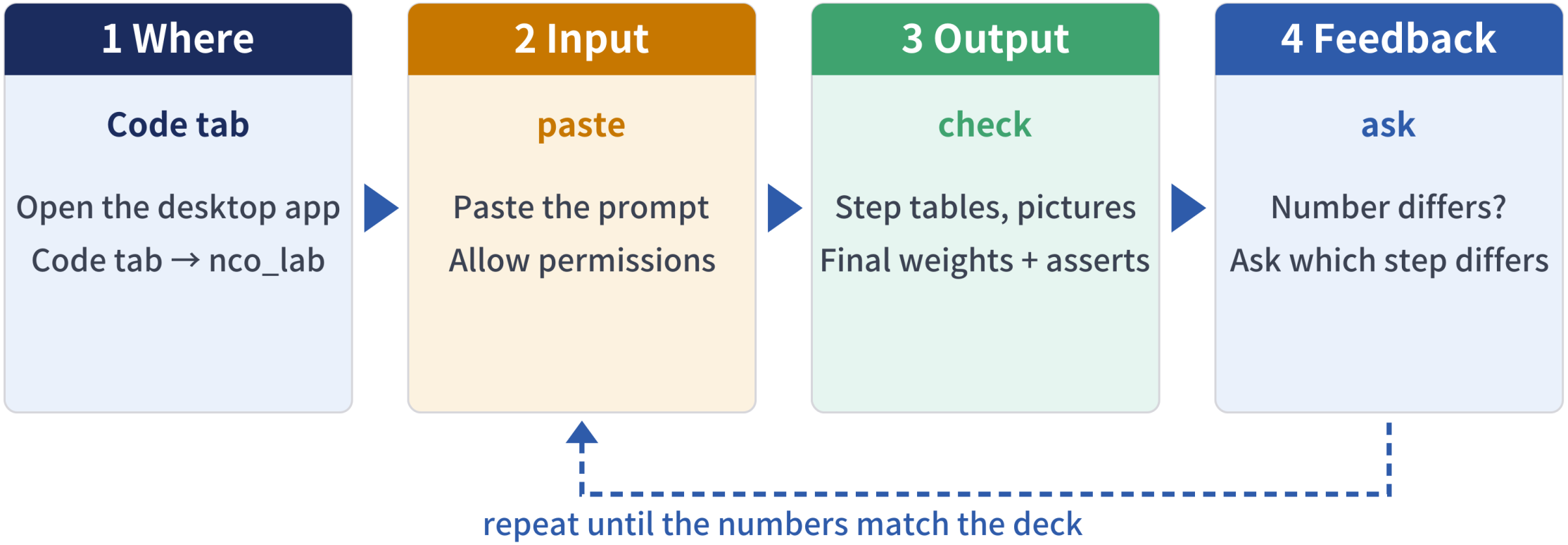
PART 5 · BUILD

Rebuilding It with Claude Code

Paste prompts into the Code tab of the Claude desktop app and check the results.

Students paste the prompts and only check the results

Claude Code installs packages, runs the code and fixes errors; students never edit code.



Prompt 1 rebuilds the four-asset example step by step

Claude Code checks itself against the deck's numbers with assert statements.

[Input] Prompt · Claude Code (Code tab)

```
In the folder nco_lab, rebuild the four-asset NCO
example in Python.
1) Vols A 0.15, B 0.20, C 0.08, D 0.12. Correlations:
   A-B 0.5, C-D 0.4, every stock-bond pair 0.2.
2) ONC: K-means on sqrt((1-corr)/2) for K=2,3;
   print silhouettes and mean/std scores.
3) Min-variance inside each cluster, the matrix W,
   W'ΣW, then min-variance across clusters.
4) Assert final weights 0.112, 0.034, 0.687, 0.167
   and compare with full min-variance on all four.
Install packages and fix errors yourself.
```

What to check

- Scores 4.6 vs 1.0
- Cluster weights 0.769, 0.805
- ω = 0.146, 0.854
- Four weights, sum 1

If the asserts pass, the code and the hand calculation agree

Prompt 2 tests NCO, HRP and MVO on real ETFs

Claude Code also downloads the data; the prompt rules out any look-ahead.

[Input] Prompt · Claude Code (Code tab)

```
Download daily adjusted prices since 2015 with
yfinance for 14 US ETFs: SPY QQQ IWM EFA EEM VNQ
TLT IEF SHY LQD HYG TIP GLD DBC.
1) Each month end, use the past 3 years: denoise the
   correlation (Marchenko-Pastur), run ONC, then NCO.
   Also compute HRP and plain minimum variance.
2) Score only with next month's returns.
3) Table: volatility, max drawdown, turnover, number
   of short or zero weights, for all three methods.
4) Plot the chosen K over time.
```

When reading

- Is NCO's turnover lower?
- Does K stay stable?
- Where does MVO go short?
- Odd? Ask which month

Check whether NCO's stability holds on real data and other periods

NCO optimizes small, then optimizes across

1

Step 1: denoise and cluster

Flatten eigenvalues below λ+, then let ONC choose K with silhouette scores.

2

Steps 2–4: inside, shrink, across

Minimum variance per cluster, W′ΣW, then minimum variance on K synthetic assets.

3

Result: smaller errors, same optimizer

Final weights 0.112, 0.034, 0.687, 0.167; in simulation, about half MVO's weight error.

Next: run Prompt 2 and compare NCO with HRP on real data